

ass4

March 2, 2025

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[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
from sklearn.metrics import r2_score
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[4]: china_gdp = pd.read_csv("C:/Users/cool/Desktop/ML ass/ass4/china_gdp.csv")
print(china_gdp.head())
```

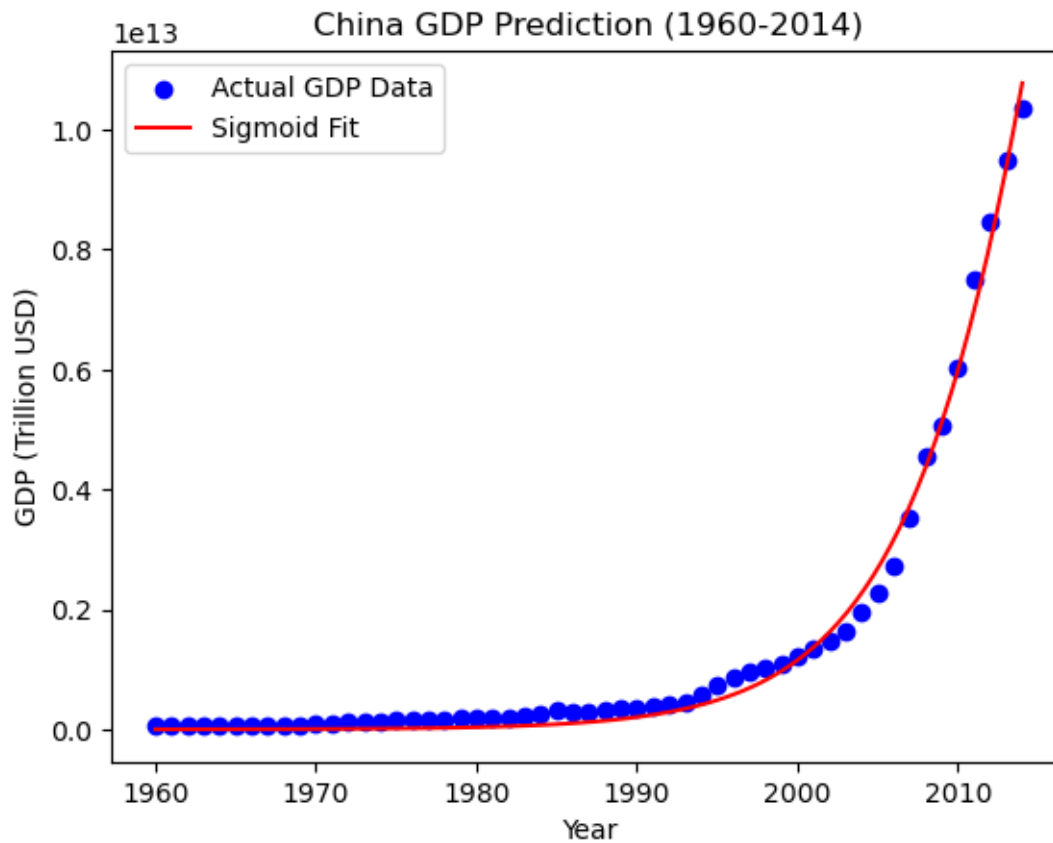
	Year	Value
0	1960	5.918412e+10
1	1961	4.955705e+10
2	1962	4.668518e+10
3	1963	5.009730e+10
4	1964	5.906225e+10

```
[6]: def sigmoid(x, L, k, x0):
return L / (1 + np.exp(-k * (x - x0)))
```

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[8]: X = china_gdp["Year"].values
y = china_gdp["Value"].values
popt, _ = curve_fit(sigmoid, X, y, p0=[max(y), 0.1, 2000]) # Initial guesses
for L, k, t0
L_opt, k_opt, x0_opt = popt
print(f"Optimized Parameters: L={L_opt:.4f}, k={k_opt:.4f}, t0={x0_opt:.4f}")
```

Optimized Parameters: L=49984395214204.3906, k=0.1752, t0=2021.3700

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[10]: x_range = np.linspace(min(X), max(X), 100)
y_pred = sigmoid(x_range, L_opt, k_opt, x0_opt)
plt.scatter(X, y, label="Actual GDP Data", color="blue")
plt.plot(x_range, y_pred, label="Sigmoid Fit", color="red")
plt.xlabel("Year")
plt.ylabel("GDP (Trillion USD)")
plt.title("China GDP Prediction (1960-2014)")
plt.legend()
plt.show()
```



```
[12]: y_pred_final = sigmoid(X, L_opt, k_opt, x0_opt)
      r2 = r2_score(y, y_pred_final)
      print(f"R2 Score: {r2:.4f}")
```

R² Score: 0.9938

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[ ]:
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